

A candidate proof of Conjecture 5 in On the difference between the Szeged and Wiener index

github.com/graph-theory-AI — machine-generated candidate writeup

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Abstract

Every 2-connected graph outside the three exceptional families satisfies $\eta(G) \geq \min\{2n, 3n-10\}$, proving the conjecture for $n \geq 10$. This candidate proof was generated by GPT-6 Astra in a single model pass for the Graph Theory AI project. It was selected from the campaign because the model marked it `would_publish: true`; it has not been checked by the adversarial referee, a human mathematician, or a novelty search.

1 The problem and the claimed result

This document concerns [Conjecture 5](#) from *On the difference between the Szeged and Wiener index*.

1.1 Problem statement

Conjecture. Let G be a 2-connected graph of order $n \geq 10$ not isomorphic to K_n , K_n^2 , or K_n^{n-2} . Then $\eta(G) \geq 2n$.

1.2 Catalog context

The authors note that while $\eta(K_n) = 0$ and $\eta(K_n^2) = \eta(K_n^{n-2}) = 2n - 6$, computer search found only finitely many 2-connected graphs with $\eta(G) < 2n$ and none on 10 or more vertices, motivating this stronger bound beyond the three exceptional families.

1.3 Claimed result

Model verdict: proved (confidence: high).

Every 2-connected graph outside the three exceptional families satisfies $\eta(G) \geq \min\{2n, 3n-10\}$, proving the conjecture for $n \geq 10$.

The model's own caveat: The proof is self-contained; novelty and current literature status have not been independently checked.

2 Model writeup

The text below is the model output verbatim, apart from moving its machine-readable verdict block into the summary above and shifting Markdown heading levels for this document. Mechanical TeX defects and equation tags were normalized where needed for compilation.

2.1 1. Statement

All graphs below are finite and simple. Write K_n^t for a clique of order $n - 1$ together with an additional vertex adjacent to exactly t clique vertices.

I prove the following stronger statement.

2.1.1 Theorem

Let G be a 2-connected graph of order $n \geq 4$, with

$$G \not\cong K_n, K_n^2, K_n^{n-2}.$$

Then

$$\boxed{\eta(G) \geq \min\{2n, 3n - 10\}.} \quad (1)$$

Consequently, if $n \geq 10$, then $\eta(G) \geq 2n$.

The universal-vertex argument in the supplied attempt is correct; a verification is included below. The additional ingredient is a reduction for graphs without universal vertices. It is important that this reduction produces a dominated vertex **whose deletion remains 2-connected**.

2.2 2. Good edges and dominated-vertex deletion

For an edge uv , let $n_{uv}(u)$ denote the number of vertices closer to u than to v . Thus

$$Sz(G) = \sum_{uv \in E(G)} n_{uv}(u)n_{uv}(v), \quad W(G) = \sum_{\{a,b\} \subseteq V(G)} d(a,b).$$

An edge uv is **good** for $\{a, b\}$ if, after possibly exchanging u, v ,

$$d(a, u) < d(a, v), \quad d(b, v) < d(b, u).$$

Let $g_G(a, b)$ be the number of good edges, and put

$$D_G(a, b) = g_G(a, b) - d_G(a, b), \quad c_G(a) = \sum_{b \neq a} D_G(a, b).$$

Every edge of every shortest a - b path is good for $\{a, b\}$. In particular,

$$D_G(a, b) \geq 0.$$

Double counting gives

$$\boxed{\eta(G) = \sum_{\{a,b\}} D_G(a, b) = \frac{1}{2} \sum_a c_G(a).} \quad (2)$$

We shall repeatedly use the following elementary deletion facts.

2.2.1 Lemma 2.1

Suppose that G is 2-connected, $|V(G)| \geq 4$, and

$$N_G[u] \subseteq N_G[v] \quad (u \neq v).$$

Put $H = G - u$.

1. H is isometric in G .
2. The only possible cut vertex of H is v .
3. If $q_G(u)$ is the number of incidences

$$(\{a, b\}, e)$$

where $a, b \in V(H)$ and e is an edge incident with u that is good for $\{a, b\}$, then

$$\eta(G) - \eta(H) = c_G(u) + q_G(u). \quad (3)$$

4. If u is simplicial, then $q_G(u) = 0$. If u is not simplicial, then $q_G(u) \geq 2$.

2.2.1.1 Proof Any portion $x - u - y$ of a path between vertices of H can be replaced by a walk through v of length at most two. This proves that deleting u preserves all distances between the remaining vertices.

Suppose $w \neq v$ were a cut vertex of H . All vertices of $N_G(u) \setminus \{w\}$ lie in the component of $H - w$ containing v , since they are adjacent to v . Adding u therefore cannot connect the components of $H - w$, contradicting the connectivity of $G - w$. This proves the second assertion.

Because distances in H are unchanged, an edge of H is good for an old pair in G exactly when it is good for that pair in H . The additional contributions are precisely the pairs containing u , and the incidences counted by $q_G(u)$. Hence (3).

If u is simplicial, then for every $x \in N(u)$ and every $a \neq u$,

$$d(a, x) \leq d(a, u).$$

Indeed, take the last neighbor of u on a shortest a - u path and use that $N(u)$ is a clique. Thus no vertex other than u is closer to u than to x , and ux cannot be good for a pair avoiding u .

Finally, if u is not simplicial, choose nonadjacent $x, y \in N(u)$. Both xu and uy are good for $\{x, y\}$, giving $q_G(u) \geq 2$. $\square$

We also use the standard consequence that deleting a simplicial vertex from a 2-connected graph of order at least four preserves 2-connectivity. For completeness, if deletion created a cut vertex w , all remaining neighbors of the simplicial vertex would lie in one component after deleting w , so restoring it could not reconnect the graph.

2.3 3. The structure of a vertex with contribution at most three

The next lemma is the main structural step.

2.3.1 Lemma 3.1 — Low-contribution structure

Let G be 2-connected, and let a be a non-universal vertex satisfying $c_G(a) \leq 3$. Put $A = N_G(a)$.

Every vertex has distance at most two from a , and one of the following descriptions holds.

2.3.1.1 Type I There are adjacent vertices $p, q \in A$ and a vertex $z \notin N[a]$ such that

$$V(G) \setminus N[a] = \{z\} \dot{\cup} X, \quad N(z) \cap A = \{p, q\}.$$

Either $X = \emptyset$, or:

- $G[X]$ is connected;
- every vertex of X has neighborhood in A exactly $\{p\}$;
- there is exactly one edge between z and X .

2.3.1.2 Type II There are distinct $p, q \in A$ and nonempty sets X, Y such that

$$V(G) \setminus N[a] = X \dot{\cup} Y,$$

where:

- $G[X]$ and $G[Y]$ are connected;
- every vertex of X has neighborhood in A exactly $\{p\}$;
- every vertex of Y has neighborhood in A exactly $\{q\}$;
- there is exactly one edge xy between X and Y .

The edges inside A , and inside the indicated connected sets, are otherwise unrestricted.

Moreover, in any 2-connected graph,

$$a \text{ non-universal} \implies c_G(a) \geq 2. \quad (4)$$

2.3.1.3 Proof Use distance layers L_i from a . A neighbor of a vertex in the preceding layer will be called a **predecessor**.

Let M be the set of vertices having more than one shortest path from a . If $b \in M$, the union of two distinct shortest a - b paths contains at least $d(a, b) + 2$ edges, all good for $\{a, b\}$. Hence

$$D_G(a, b) \geq 2. \quad (5)$$

It follows that $|M| \leq 1$.

We first record another useful observation. Suppose $x, y \in L_i$ have unique shortest paths from a , their predecessors are different, and $xy \in E(G)$. If $p(y)$ denotes the predecessor of y , then

$$d(x, y) = 1, \quad d(x, p(y)) = 2.$$

Thus $p(y)y$ is good for $\{a, x\}$, and it does not belong to the unique shortest a - x path. The analogous assertion holds with x, y exchanged.

For a fixed x , different such neighbors y give different extra good edges.

2.3.2 Case 1: $M = \emptyset$

The edges between consecutive layers form a rooted tree T : every non-root vertex has exactly one predecessor.

Let F be the set of same-layer edges whose endpoints have different predecessors. The preceding observation gives

$$c_G(a) \geq 2|F|. \quad (6)$$

Thus $|F| \leq 1$.

If $F = \emptyset$, all non-tree edges join siblings. In such a graph, the proper descendants of any non-root tree vertex have no neighbors outside that subtree except its root. Consequently, a vertex at distance at least two from a would force a cut vertex. Since G is 2-connected, a would be universal, contrary to assumption.

Therefore $F = \{xy\}$, say $x, y \in L_i$, with distinct predecessors p, q .

We claim that p, q have the same predecessor. Otherwise p, q are nonadjacent, since pq would be a second edge of F . They also have no common neighbor:

- a common neighbor in L_{i-2} contradicts their distinct predecessors;
- a common neighbor in L_{i-1} would produce another edge of F ;
- a common neighbor in L_i would have two shortest paths from a .

Hence

$$d(p, q) = 3$$

via $p - x - y - q$, while $d(p, y) = d(q, x) = 2$. Therefore qy is an extra good edge for $\{a, p\}$, and px is an extra good edge for $\{a, q\}$. Together with the extra contributions at x and y , this gives $c_G(a) \geq 4$, a contradiction.

Let w be the common predecessor of p, q . Apart from sibling edges, the only non-tree edge is xy , which lies entirely among proper descendants of w . If $w \neq a$, those descendants are separated from a by w , contradicting 2-connectivity. Thus $w = a$, and $i = 2$.

There can be no vertex in L_3 or a later layer. Indeed, the proper descendants of its ancestor in L_2 would have no exit except through that ancestor: the sole exceptional edge xy has both endpoints in L_2 .

Every vertex of L_2 therefore has a unique neighbor in A . The only edge between different predecessor classes is xy . A class other than those of p, q , or a component of the p -class not containing x , would be separated by its predecessor. The analogous statement holds for q . This gives Type II.

2.3.3 Case 2: $M = \{z\}$

The vertex z has at least two predecessors; otherwise its unique predecessor would also belong to M . It has no neighbor in the next layer, since such a neighbor would also belong to M .

If z had $r \geq 3$ predecessors, the last two layers of its shortest paths would contain at least $2r$ good edges, only two of which belong to one fixed shortest path. Hence

$$D_G(a, z) \geq 2r - 2 \geq 4,$$

which is impossible. Thus z has exactly two predecessors, say p, q .

If $pq \notin E(G)$, then qz is an extra good edge for $\{a, p\}$, and pz is one for $\{a, q\}$. Together with $D_G(a, z) \geq 2$, this again gives $c_G(a) \geq 4$. Hence $pq \in E(G)$.

There cannot be a same-layer edge between two uniquely reached vertices with different predecessors: it would contribute at least one at each endpoint in addition to the contribution of at least two at z . In particular, p, q have the same predecessor, say w .

Consider a same-layer neighbor t of z . Its unique predecessor must be p or q . Otherwise both pz and qz are extra good edges for $\{a, t\}$, giving $D_G(a, t) \geq 2$. If its predecessor is p or q , it still contributes at least one. Thus z has at most one same-layer neighbor.

Choose a breadth-first tree by making p the predecessor of z . All non-tree edges are sibling edges, except qz and the possible same-layer edge incident with z . These exceptional edges lie entirely

among proper descendants of w . As in Case 1, 2-connectivity forces $w = a$. Thus $z \in L_2$. The same subtree-separation argument rules out all later layers.

If z has no same-layer neighbor, every other predecessor class in L_2 would be separated by its predecessor, so $L_2 = \{z\}$.

Otherwise let t be its sole same-layer neighbor, and exchange p, q if necessary so that p is the predecessor of t . All remaining vertices of L_2 must have predecessor p , and their induced graph must be connected and contain t ; any other component would be separated by its predecessor. This is Type I.

Finally, if $c_G(a) \leq 1$, then (5) forces $M = \emptyset$, and (6) forces $F = \emptyset$. The argument in Case 1 then makes a universal. This proves (4). $\square$

2.4 4. Consequences of the low-contribution structure

2.4.1 Lemma 4.1 — A 2-connected deletion exists

Let G be 2-connected with no universal vertex. If some vertex has contribution at most three, then either $G \cong C_5$, or there are distinct vertices u, v such that

$$N[u] \subseteq N[v] \quad \text{and} \quad G - u \text{ is 2-connected.} \quad (7)$$

2.4.1.1 Proof Apply Lemma 3.1.

2.4.1.2 Type I If $X = \emptyset$, then z is simplicial, with adjacent neighbors p, q , so deleting z preserves 2-connectivity.

Suppose $X \neq \emptyset$, and let zt be its sole edge to z . Every vertex of X is dominated by p .

If $X = \{t\}$, then t is simplicial, with adjacent neighbors p, z , and can be deleted.

Otherwise choose $u \in X \setminus \{t\}$ such that $G[X] - u$ remains connected. Such a vertex exists by taking a leaf other than t in a spanning tree of $G[X]$.

The graph $G - p$ is connected, and X is attached to its complement there only through tz . Deleting u preserves this connectivity. Thus $G - \{u, p\}$ is connected. Since u is dominated by p , Lemma 2.1 shows that $G - u$ is 2-connected.

2.4.1.3 Type II If $|X| \geq 2$, choose a non-cut vertex $u \neq x$ of $G[X]$. It is dominated by p . In $G - p$, the set X is attached to its complement only by xy , and deleting u preserves connectivity. Lemma 2.1 again gives a 2-connected deletion. The case $|Y| \geq 2$ is symmetric.

We may therefore assume $X = \{x\}$ and $Y = \{y\}$.

If $A = \{p, q\}$, then:

- if $pq \notin E(G)$, the graph is C_5 ;
- if $pq \in E(G)$, the vertex a is simplicial and can be deleted.

Finally suppose $|A| \geq 3$. The graph

$$R = G[A] + pq$$

is connected, because $G - a$ is connected and the path $p - x - y - q$ connects p to q .

Choose a spanning tree of R containing pq . Since this tree has at least three vertices, it has a leaf $u \notin \{p, q\}$. Thus $R - u$ is connected. The vertex u is dominated by a , and replacing the added edge pq by $p - x - y - q$ shows that $G - \{a, u\}$ is connected. Lemma 2.1 proves that $G - u$ is 2-connected. $\square$

The next lemma prevents low-increment simplicial deletions in a counterexample without a universal vertex.

2.4.2 Lemma 4.2 — Low-contribution simplicial vertices

Let G be 2-connected and have no universal vertex. If a is simplicial and $c_G(a) \leq 3$, then

$$\eta(G) \geq 2|V(G)|. \quad (8)$$

2.4.2.1 Proof In Lemma 3.1, $A = N(a)$ is now a clique.

Type I is impossible: the vertex p is adjacent to a , all of A , z , and all of X , so it is universal.

Thus Type II holds. Put

$$K = \{a\} \cup A.$$

The vertices p, x, y, q induce the cycle

$$p - x - y - q - p.$$

It is isometric, so the sum of D_G over pairs within these four vertices is at least

$$\eta(C_4) = 8. \quad (9)$$

We show that every other vertex supplies at least two additional units, using only pairs with these four vertices.

2.4.2.2 Vertices in $K \setminus \{p, q\}$ For $b \in K \setminus \{p, q\}$, the path $b - p - x$ is shortest, and qy is an additional good edge for $\{b, x\}$. Hence

$$D_G(b, x) \geq 1.$$

Symmetrically,

$$D_G(b, y) \geq 1.$$

2.4.2.3 Vertices in $X \setminus \{x\}$ Let $b \in X \setminus \{x\}$.

If $bx \in E(G)$, then:

- $b - p - q$ is shortest and xy is an additional good edge for $\{b, q\}$;
- $b - x - y$ is shortest and pq is an additional good edge for $\{b, y\}$.

Thus $D_G(b, q) + D_G(b, y) \geq 2$.

If $bx \notin E(G)$, then $d(b, y) = 3$, and both

$$b - p - q - y, \quad b - p - x - y$$

are shortest paths. Their union has five edges, so

$$D_G(b, y) \geq 2.$$

The argument for $Y \setminus \{y\}$ is symmetric.

These counted pairs are distinct: each has one endpoint outside the four-cycle and one inside it. All uncounted contributions are nonnegative. Therefore

$$\eta(G) \geq 8 + 2(|V(G)| - 4) = 2|V(G)|.$$

□

Combining the preceding lemmas gives the reduction needed for induction.

2.4.3 Proposition 4.3 — Reduction with increment at least four

Let G be 2-connected, with no universal vertex, and suppose

$$\eta(G) < 2|V(G)|.$$

Then either $G \cong C_5$, or there is a vertex u dominated by another vertex such that $H = G - u$ is 2-connected and

$$\boxed{\eta(G) - \eta(H) \geq 4.} \quad (10)$$

2.4.3.1 Proof By (2), some vertex has contribution at most three. Unless $G = C_5$, Lemma 4.1 supplies the required dominated vertex u with a 2-connected deletion.

If u is not simplicial, then Lemma 3.1 gives $c_G(u) \geq 2$, and Lemma 2.1 gives $q_G(u) \geq 2$. Hence the increment is at least four.

If u is simplicial, Lemma 4.2 and the hypothesis $\eta(G) < 2|V(G)|$ force $c_G(u) \geq 4$. Equation (3) again gives the result. $\square$

2.5 5. Graphs with a universal vertex

This section verifies the relevant result from the supplied attempt.

For a graph Q , let $p_3(Q), p_4(Q), c_4(Q)$ count induced copies of P_3, P_4, C_4 , respectively.

2.5.1 Lemma 5.1

If $\text{diam}(G) \leq 2$, then

$$\eta(G) = 2(p_3(G) - \overline{m}(G)) + p_4(G) + 4c_4(G), \quad (11)$$

where $\overline{m}(G)$ is the number of nonedges.

2.5.1.1 Proof For a nonedge ab , every common neighbor gives two good edges. Every other good edge is the middle edge of an induced P_4 with endpoints a, b . Thus

$$D_G(a, b) = 2(|N(a) \cap N(b)| - 1) + \#\{\text{induced } P_4 \text{'s with endpoints } a, b\}.$$

For an edge ab , every good edge other than ab is the edge opposite ab in an induced C_4 . Summing over pairs counts each induced C_4 four times. Summing common-neighbor counts over nonedges counts $p_3(G)$. This proves (11). $\square$

In particular, for a cone $G = K_1 \vee Q$,

$$\boxed{\eta(G) = 2p_3(Q) + p_4(Q) + 4c_4(Q).} \quad (12)$$

Indeed, the apex creates one additional induced P_3 for each nonedge of Q , and belongs to no induced P_4 or C_4 .

2.5.2 Lemma 5.2

Let Q be a connected, noncomplete graph of order m . Unless Q is $K_m - e$ or a clique K_{m-1} with one pendant vertex,

$$2p_3(Q) + p_4(Q) + 4c_4(Q) \geq 3m - 7. \quad (13)$$

For each of the two exceptional graphs, the left side equals $2m - 4$.

2.5.2.1 Proof Write

$$F(Q) = 2p_3(Q) + p_4(Q) + 4c_4(Q).$$

Choose a maximum clique C , of order $c \geq 2$. Let L_i now denote distance layers from C , and put

$$\ell = |L_1|, \quad r = \sum_{i \geq 2} |L_i|.$$

Thus $m = c + \ell + r$, and $\ell \geq 1$.

For $x \in L_1$, set $a_x = |N(x) \cap C|$. Maximality of C gives

$$1 \leq a_x \leq c - 1.$$

The vertex x , one neighbor in C , and one nonneighbor in C give

$$a_x(c - a_x) \geq c - 1$$

induced P_3 's.

Every vertex in L_i , $i \geq 2$, supplies another induced P_3 , using the first three vertices of a shortest path to C . These copies are distinct, by their unique vertex of maximum layer. Consequently,

$$p_3(Q) \geq (c - 1)\ell + r. \quad (14)$$

Likewise,

$$p_4(Q) \geq r. \quad (15)$$

For a vertex at distance at least three, take four vertices of a shortest path. For $x \in L_2$, take $x - y - v$ with $v \in C$, and choose $w \in C \setminus N(y)$; then $x - y - v - w$ is induced. Again the maximum-layer vertex distinguishes the copies.

2.5.2.2 Case 1: $c \geq 3$ and $\ell \geq 2$ Equations (14)–(15) give

$$F(Q) \geq 2(c - 1)\ell + 3r.$$

The difference from $3m - 7$ is

$$(2c - 5)\ell - 3c + 7 \geq c - 3 \geq 0.$$

2.5.2.3 Case 2: $c \geq 3$ and $\ell = 1$ Write $L_1 = \{x\}$, $a = |N(x) \cap C|$, and $A = a(c - a) \geq c - 1$.

If $r = 0$, then $F(Q) = 2A$. The cases $a = 1$ and $a = c - 1$ are the two exceptions. Otherwise $2 \leq a \leq c - 2$, and

$$F(Q) \geq 4(c - 2) \geq 3c - 4 = 3m - 7.$$

Suppose $r > 0$, and write

$$s = |L_2|, \quad t = \sum_{i \geq 3} |L_i|.$$

Every vertex of L_2 is adjacent to x . It supplies A induced paths $y - x - v - w$, where $v \in N(x) \cap C$ and $w \in C \setminus N(x)$. Hence

$$p_3(Q) \geq A + s + t, \quad p_4(Q) \geq As + t.$$

Therefore

$$F(Q) \geq 2A + (A + 2)s + 3t.$$

Subtracting $3m - 7 = 3c + 3s + 3t - 4$, and using $s \geq 1$, gives

$$2A + (A - 1)s - 3c + 4 \geq 2(c - 1) + (c - 2) - 3c + 4 = 0.$$

2.5.2.4 Case 3: $c = 2$ Write $C = \{u, v\}$. The graph is triangle-free. Partition L_1 into

$$A = N(u) \cap L_1, \quad B = N(v) \cap L_1,$$

with sizes α, β , so $\ell = \alpha + \beta$. Let e be the number of edges between A and B .

Besides the copies in (14), pairs in A and in B give

$$\binom{\alpha}{2} + \binom{\beta}{2}$$

additional induced P_3 's. A nonedge between A, B gives an induced P_4 through u, v , and an edge between them gives an induced C_4 . These copies are disjoint from those counted in (15). Thus

$$\begin{aligned} F(Q) &\geq 2 \left(\ell + r + \binom{\alpha}{2} + \binom{\beta}{2} \right) + r + (\alpha\beta - e) + 4e \\ &= 2\ell + 3r + \alpha(\alpha - 1) + \beta(\beta - 1) + \alpha\beta + 3e. \end{aligned}$$

For nonnegative integers α, β with $\alpha + \beta \geq 1$,

$$\alpha(\alpha - 1) + \beta(\beta - 1) + \alpha\beta \geq \alpha + \beta - 1.$$

If both are positive, this follows by bounding the first two terms by $\alpha - 1, \beta - 1$, and $\alpha\beta$ by 1; if one is zero it is immediate. Hence

$$F(Q) \geq 3\ell + 3r - 1 = 3m - 7.$$

Finally, each exceptional graph has $m - 2$ induced P_3 's and no induced P_4 or C_4 , giving $F(Q) = 2m - 4$. $\square$

2.5.3 Proposition 5.3

Let G be 2-connected of order n , with a universal vertex. If

$$G \not\cong K_n, K_n^2, K_n^{n-2},$$

then

$$\boxed{\eta(G) \geq 3n - 10.} \quad (16)$$

2.5.3.1 Proof Write $G = K_1 \vee Q$. The graph Q is connected, since deleting the universal vertex preserves connectivity.

If Q is complete, $G = K_n$. The two exceptions of Lemma 5.2 give, respectively, K_n^{n-2} and K_n^2 . Otherwise (12)–(13), with $m = n - 1$, give

$$\eta(G) \geq 3(n - 1) - 7 = 3n - 10.$$

$\square$

2.6 6. Proof of the theorem

First, formula (11) gives

$$\boxed{\eta(K_m^t) = 2(m - 1 - t)(t - 1).} \quad (17)$$

Indeed, there are $m - 1 - t$ nonedges, each with t common neighbors, and no induced P_4 or C_4 . In particular,

$$\eta(K_m^2) = \eta(K_m^{m-2}) = 2m - 6. \quad (18)$$

Put

$$f(n) = \min\{2n, 3n - 10\}.$$

We prove (1) by induction on n .

For $n = 4$, the only admissible graph is C_4 , and $\eta(C_4) = 8 \geq f(4)$.

Now let $n \geq 5$, and suppose the assertion holds at smaller orders.

If G has a universal vertex, Proposition 5.3 gives the result. We may therefore assume that G has no universal vertex.

If $\eta(G) \geq 2n$, there is nothing to prove. If $G = C_5$, then

$$\eta(G) = 20 - 15 = 5 = f(5).$$

Otherwise Proposition 4.3 supplies a dominated vertex u such that

$$H = G - u$$

is 2-connected and

$$\eta(G) \geq \eta(H) + 4. \quad (19)$$

Write $m = n - 1$.

2.6.1 Case 1: H is not exceptional

By induction,

$$\eta(G) \geq f(n - 1) + 4.$$

But

$$f(n - 1) + 4 = \min\{2n + 2, 3n - 9\} \geq \min\{2n, 3n - 10\} = f(n).$$

So this case is finished.

It remains to check every possible exceptional predecessor.

2.6.2 Case 2: $H \cong K_m$

Every neighbor of u is universal in G , contrary to our assumption that G has no universal vertex.

2.6.3 Case 3: $H \cong K_m^{m-2} = K_m - e$

Let x, y be the endpoints of its missing edge. Every vertex of $H \setminus \{x, y\}$ is universal in H . Since G has no universal vertex, none is adjacent to u .

Because G is 2-connected, $d_G(u) \geq 2$, so

$$N_G(u) = \{x, y\}.$$

But neither x nor y can dominate u , since $xy \notin E(G)$. This contradicts the choice of u .

2.6.4 Case 4: $H \cong K_m^2$

The case $m = 4$ is already included in Case 3. Otherwise write H as a clique B of order $m - 1$ and an additional vertex z with

$$N_H(z) = \{p, q\} \subseteq B.$$

Put

$$R = B \setminus \{p, q\}.$$

The vertices p, q are universal in H . Thus u is adjacent to neither of them.

We claim that u is not adjacent to z . Otherwise, since $d_G(u) \geq 2$, it also has a neighbor in R . A vertex dominating u would have to be a neighbor of u ; but z is adjacent to no vertex of R , and no vertex of R is adjacent to z . No possible dominator exists.

Consequently,

$$N_G(u) \subseteq R.$$

Let $t = |N_G(u)| \geq 2$. The vertex u is simplicial.

Each of p, q has all t vertices of $N(u)$ as common neighbors with u , so

$$D_G(u, p) \geq 2(t-1), \quad D_G(u, q) \geq 2(t-1). \quad (20)$$

Also $d_G(u, z) = 3$. For every $r \in N(u)$, the two paths

$$u - r - p - z, \quad u - r - q - z$$

are shortest. Their union over all r has $3t + 2$ edges, all good for $\{u, z\}$. Therefore

$$D_G(u, z) \geq 3t - 1. \quad (21)$$

Combining (20)–(21),

$$c_G(u) \geq 7t - 5 \geq 9.$$

By Lemma 2.1 and (18),

$$\eta(G) \geq (2m - 6) + 9 = 2n + 1.$$

This contradicts the case assumption $\eta(G) < 2n$.

All exceptional predecessors have now been covered, completing the induction and proving (1). $\square$

Since $3n - 10 \geq 2n$ exactly when $n \geq 10$, the conjecture follows.

2.7 7. Sharpness and scope

The stronger bound is sharp for every $n \geq 5$.

2.7.0.1 The $3n - 10$ branch

$$G = K_1 \vee P_{n-1},$$

equation (12) gives

$$\eta(G) = 2(n-3) + (n-4) = 3n - 10.$$

These graphs are admissible for $n \geq 5$.

2.7.0.2 The $2n$ branch Start with the cycle $p - x - y - q - p$, and enlarge its edge pq to a clique by adding $n - 4$ vertices, each adjacent to every vertex already in that clique and to neither x nor y .

Equivalently, take a clique of order $n - 2$, an additional edge xy , and exactly two cross edges forming a matching.

Each added vertex is simplicial and has contribution exactly two: its only nonzero contributions are one with x and one with y . Lemma 2.1 therefore gives

$$\eta(G) = \eta(C_4) + 2(n-4) = 2n.$$

Thus the transition at $n = 10$ is genuine.

No computational enumeration or unproved external assertion is used. In particular, the previously untreated graphs without universal vertices are covered by Proposition 4.3, and the induction explicitly handles all three exceptional predecessor families.

A Provenance and review status

UNREFEREED MODEL OUTPUT. This document was selected solely because GPT-6 Astra labelled its own result `would_publish: true` and returned `proved` or `disproved`. It has not passed the adversarial LLM referee used for the earlier Sol campaign, has not been checked by a human mathematician, and has not been checked for novelty. Treat every mathematical and bibliographic claim as unverified.

Catalog id	1602.05184__00
Catalog entry	Conjecture 5
Source corpus	arxiv
Campaign leg	selected second attempts (<code>attacks_retry</code>)
Source paper / entry	On the difference between the Szeged and Wiener index
Model verdict	proved (confidence: high)
Selection criterion	<code>would_publish: true</code> and verdict <code>proved</code> or <code>disproved</code>
Model	<code>gpt-6-astra</code> , reasoning effort <code>max</code> , mode= <code>pro</code> , flex service tier
Original artifact	<code>attacks_retry/1602.05184__00/output.md</code>
Independent review	None
arXiv	arXiv:1602.05184
Previous attempt	<code>attacks/1602.05184__00: gpt-5.6-sol verdict partial</code>
Model's caveats	The proof is self-contained; novelty and current literature status have not been independently checked.

The generated Markdown and LaTeX sources for this PDF are stored under `to_review_astra/src/1602.05184__00/`.